

Two applications of field theory. Remarks

Prologue: What is the difference between TFT and TQFT?

Mathematical Gauge theory mostly refers to classical theory. For example, the CS and YM (CS is boundary of YM, boundary of CS is CFT) via connection 1-forms. This mathematical interpretation was based on rewriting the Maxwell equations; we rewrite the four components vector fields into four components differential 1-form to adjust the calculus $\nabla \cdot, \nabla \times, \dots$, this used Hodge duality intrinsically (in physical language, we have sum in ~~form~~ the form $A_{\mu\nu} A^{\mu\nu}$, here is $A_{\mu\nu}$ vector field but $A^{\mu\nu}$ its Hodge dual, pair). Canonical quantization is solve the components (four or two) as (four or two) functions, then use $a, a^\dagger$ to quantize it into operator with (four or two) components.

Thus a differential form/vector field can act on function, doesn't mean that we should take it as ~~field~~ field operator after quantization, but they're classical. One may call the connection 1-form as "photon," it's equivalent to its quantized field operator is photon.

In these theories' Lagrangian, we may also have Feynman-like diagram, but ① Their forms have no integral over space-time; ② Not perturbative expansion, but computed concretely;

Thus it has totally different physical meaning: classically, $\times$ means the interaction between four particles, quantumly, $\times$ means the scattering process 2-to-2, ~~and~~ not complete, it needs more quantum correlations $\times \times + \dots$



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But when we consider moduli space with all such A , it's equivalent to consider the field configuration space ^{by Riemann-Hilbert, it's same to after quantization, thus study the} variational method of given Lagrangian / ^{consider field operator as its solution} Morse-type theory in ^(quantum) the ^(classical) field configuration space is subtle, we can say that we are considering $\sigma: CO, 11 \rightarrow M$ as σ -model, and this is exactly quantized theory, or some string theory ~~turn~~ turn up. Thus one may call Floer theory is quantized, ~~and~~ or as a ~~consequence~~ consequence of open string theory. Another confusing point is the path-ordered exponential, when I first study gauge theory, a physical student argued with me that it's the correlation function/path-integral as a quantized theory. But it's only the time-ordered reason, it occurs in correlation function, but it's classical: the exponential is only for parallel transport on single path, than path-integral.

In next two applications, we'll consider all TQFT, in mathematics, we always use quantized as we need "input" & "output" in scattering theory, just as Atiyah-Segal axioms. TFT is always quantized and super-symmetry, if we want ~~to~~ to talk about physics, thus it's hard to say it has any concrete physical meaning.

PART I Mirror Symmetry

The T-duality in physics is a Fourier transform $\int_{\text{Map}(\Sigma, X)} \square = \int_{\text{Map}(\Sigma, \hat{X})} \square$ applied to infinite-dim path-integral, with Σ is Riemann surface, $\text{Map}(\Sigma, X)$ is σ -model of SCFT of X . It relates to mathematical MS by localization formula: left localize topological Map to holomorphic $\text{Hol}(\Sigma, X)$, right localize to constant map $\cong \hat{X}$.

$\Rightarrow \int_{\text{Hol}(\Sigma, X)} \square = \int_{\hat{X}} \square$ ^(we had seen it in Witten's PS 10, AS index then) ok. Using different model via phase transition, via SUSY there're non-Kähler version such as ^{this is Witten's Rosetta Stone} ~~London-Gibbs~~ London-Gibbs.

Thus left is A-model, studying maps; right is B-model, studying algebraic geometry of $\hat{X}$ (as the integral can be realized as algebraic for $\hat{X}$ Kähler) (replace SCFT by SSFT)

Replace Σ by world-sheet of string, then these integrals are algebraic, as closed string $\leadsto$ L_{∞} -Algb

Open string $\leadsto$ Assoc-Algb

and these bounded theories are more interesting.

Rk. $X, \hat{X}$ are classical spaces, but integral over σ -model is quantized. the localization is taking effective theory of ~~g~~ GFT, localize $\text{Map}(\Sigma, \hat{X})$ to $\hat{X}$ turns back to classical isn't so surprising, just as low energy limit of QM is classical Newtonian Mechanics.

PART II Geometric Langlands Programs

Lafforgue has a famous snake. I'd like to first explain how they're linked:

Physics ① 4d TFTs S-duality ② 3-mfds ③ Curves over $\mathbb{F}_q$ ④ Langlands

① No problem;

② 3-mflds' cobordism is 4d TQFT;

③ ~~Algebraic~~ Algebraic Curves $\leftrightarrow$ Riemann Surfaces, this's Weil's Rosetta.

Stone, Mazur and ... took the Frobenius / $\mathbb{F}_q$ into consideration, each closed point in $\mathbb{C}/\mathbb{F}_q$ has $\neq 1$ nontrivial $\hat{\mathbb{Z}}$ generated by Frob

$\Rightarrow$ add 1-dim to 3-mfd, with S^1 -fibration;

④ $\mathbb{C}/\mathbb{F}_q \Leftrightarrow \mathbb{F}_q(t)/\mathbb{F}_q$ the function field case of Langlands, the number field case is analogous $K/\mathbb{Q}$ then, or local version $\mathbb{F}_q((t))$ and $K/\mathbb{Q}_p$, then it's arithmetic story.


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Our application of T&FT is purely algebraic here, the unramified GUP, or geometric Satake correspondence, says that the Hecke algebra $\mathbb{C}[G(\mathbb{Q}_p) \backslash G(K_p) / G(\mathbb{O}_p)]$ is a polynomial algebra / $\mathbb{C}$, i.e. commutative.

Q. Why it's equivalent to the form of GLP we know? I don't know

Consider the Hecke algebra $\mathbb{C}[H \backslash G / H] = \mathbb{C}[\cdot / H \times \cdot / H] \cong$ groupoid algebra $\cong \mathbb{C}[\text{Loc}_H(\cdot) \times \text{Loc}_H(\cdot)]$

We need a functor $(\text{Loc}_G(X) := \text{Hom}(X, \cdot / G))$

We need a functor

$$\text{Bord} \xrightarrow{\text{log}_G(-)} \text{Con}(G\text{-pd}) \xrightarrow{\text{df-1}} \text{Vect}_G$$

the category of same object with Grpd .

but morphism are correspondence / span, like Bird cat

Then the source and target advise it's a TQFT functor $Z_G(-)$,
and we can check it.

$\Rightarrow \text{Coh}(G/H) = \bigoplus_{\mathbb{Z}_H(\cdot)} \mathbb{Z}_H(\cdot) \in \text{Vect}_C$, then we can prove it:

Step 1 Dealing the tensor / fibre product, how to collect Z_H & Z_G into one TAF

Step 2 Realize the product as OPE (operator product expansion)

"If ~~we want to go further~~ We need the bipartite TQFT, and the boundary condition enhance to higher category: let $\mathcal{A} \left(\begin{array}{c} \text{diagram of a square with a diagonal line and a shaded region} \end{array} \right)^{x^d}$ $\cong \mathcal{Z}_V(x)$

$F: Z_V(X) \rightarrow Z_W(Y)$ is given by $f \in \text{Hom}(V, W)^{Y^d}$

ie. span 1:

Let $V=W=\mathbb{C}[G/H] \Rightarrow$ the interface is $\mathbb{C}[H]$

Then consider two interfaces case:

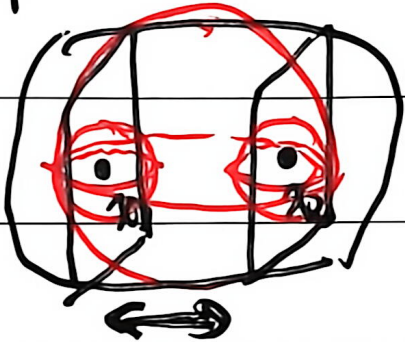
now let "topological" works:

$\Rightarrow$ two interfaces composite $\Rightarrow \text{CHG/H} \otimes \text{CHG/H} \rightarrow \text{CHG/H}$

Eckmann-Hilton argument

and this OPE is obviously commutative, done.

Step 2 How to composite of interface works?

Two interfaces  and x_1, x_2 on it, their local operators depends on the ball near it

find a larger ball containing it (due to topological reason, there always $\exists$) $\Rightarrow Z(S^{n-1}) \otimes Z(S^{n-1}) \rightarrow Z(S^{n-1})$ deformed to local operators' OPE topologically 

Rk. S-dual & T-dual are both Fourier transform in some physical configuration spaces, and have clear physical meaning, i.e. intertwine x and p . How to understand the Fourier transform?